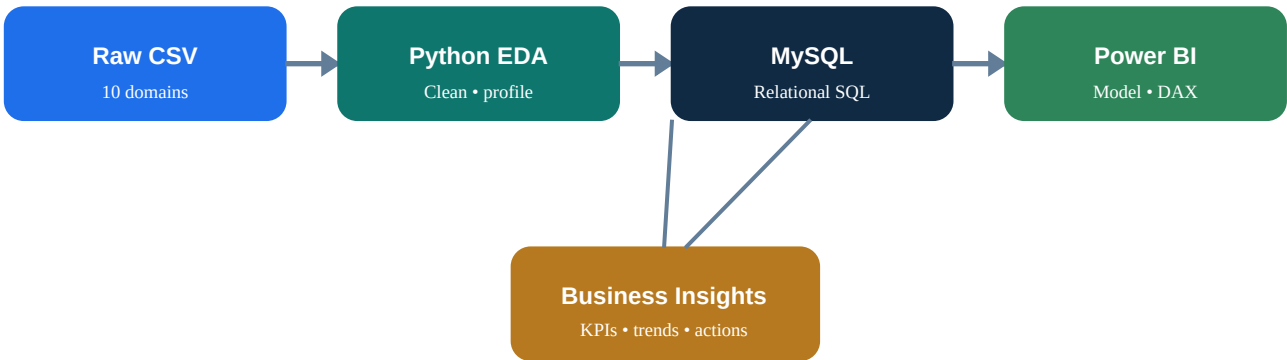


BANKING TRANSACTIONS ANALYTICS

Professional Data Analyst Case Study

Python • MySQL • Power BI • DAX

60K	95K	2.0M	■12.10B
Customers	Accounts	Core transactions	Core transaction value



Portfolio positioning

An end-to-end banking analytics project showing how raw multi-domain data can be transformed into validated analysis, interactive reporting, quantified insights and business recommendations.

Source basis: supplied project report and SQL source. Quantified figures are limited to values verifiable from those artifacts.

01 — Executive Summary

The project combines Python exploratory analysis, MySQL relational analysis and Power BI reporting across customers, accounts, branches, transactions, cards, loans, loan payments, employees and support.

3.0M	65K	22K	600K
Card transactions	Cards	Loans	Loan payments

Business challenge

Management needs a consolidated view of customer behavior, account activity, transaction patterns, card lifecycle, repayment behavior and support indicators.

Key findings

Customer concentration	Maharashtra 19.97%; Madhya Pradesh 13.43%.
Account activity	84.96% active; 10.03% dormant; 5.02% closed.
Transactions	■12.10B core transaction value; ■6,048.38 average.
Digital channel	Mobile App: 333,776 core transactions.
Card lifecycle	63.36% classified as expired using notebook expiry logic.
Repayment	12.0063% late-payment flag rate.

Analyst takeaway

The project demonstrates raw data → quality checks → EDA → SQL analysis → Power BI modeling → dashboard storytelling → business recommendations.

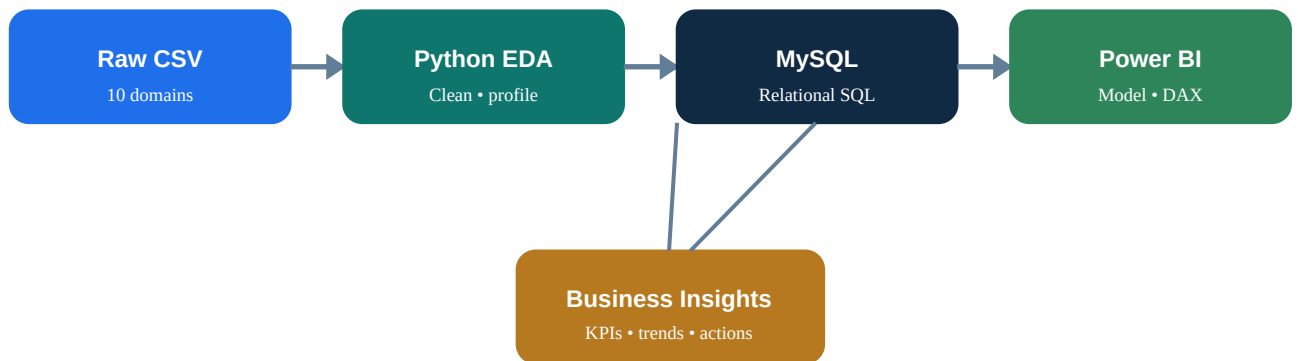
02 — Business Problem & Analytical Framework

The objective was to build a structured analytical layer that answers business questions across multiple banking domains, rather than simply creating charts.

Business questions

- Who are the customers and where are they concentrated?
- How active are accounts and which account types dominate?
- Which transaction types and channels drive activity and value?
- What does the card portfolio lifecycle look like?
- What repayment and late-payment patterns are visible?
- How can credit and support information be investigated together?

End-to-end architecture



Analytical layers

Layer	Purpose	Output
Python	Profile, clean and explore	EDA, trends, QA
MySQL	Relationships and business questions	SQL metrics
Power BI	Model, calculate, visualize	Dashboards and KPIs
Business	Translate findings to action	Insights and recommendations

03 — Data Landscape & Quality

The project spans ten banking domains and uses a relational layer connecting customers/accounts/branches, accounts/transactions, cards/card transactions, loans/loan payments, employees/branches and support/customers.

Domain	Rows	Purpose
Customers	60,000	Profile, geography, income, credit score
Accounts	95,000	Status, type, balances, branch
Branches	150	Branch footprint
Transactions	2,000,000	Volume, value, type, channel
Cards	65,000	Portfolio and lifecycle
Card Transactions	3,000,000	Card spending
Loans	22,000	Loan portfolio
Loan Payments	600,000	Repayment and late payments
Employees	1,800	Branch staffing
Support	25,000	Ticket and service analysis

Quality checks

The workflow includes structure/type inspection, duplicate and null checks, identifier validation, date/numeric standardization and feature engineering for age groups, credit-status categories, transaction month/year and card status.

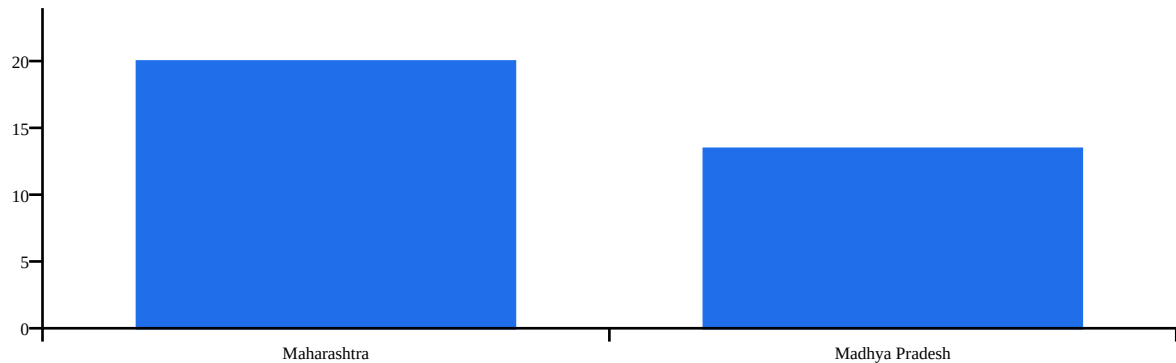
QA note

The supplied documentation flags exploratory SQL queries that need review before production use, including missing ORDER BY clauses, questionable aliases and potential logic issues. Power BI KPIs should be reconciled against one validated baseline.

04 — Customer & Account Analysis

19.97%	13.43%	84.96%	50.06%
Maharashtra share	Madhya Pradesh share	Active accounts	Savings share

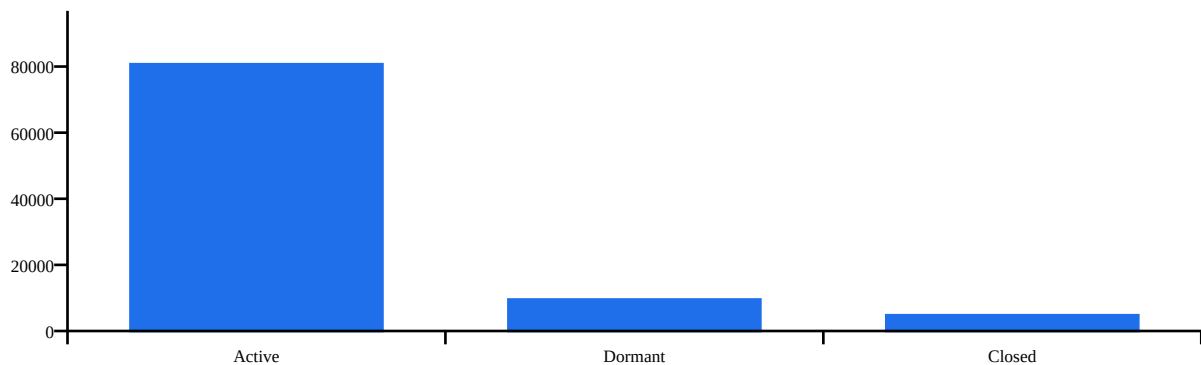
Customer concentration — leading states



Maharashtra contains 11,981 customers (19.97%), followed by Madhya Pradesh with 8,059 (13.43%).

Account activity

Account status distribution



The portfolio contains 80,707 active, 9,526 dormant and 4,767 closed accounts. Savings accounts are the largest type with 47,557 accounts (50.06%).

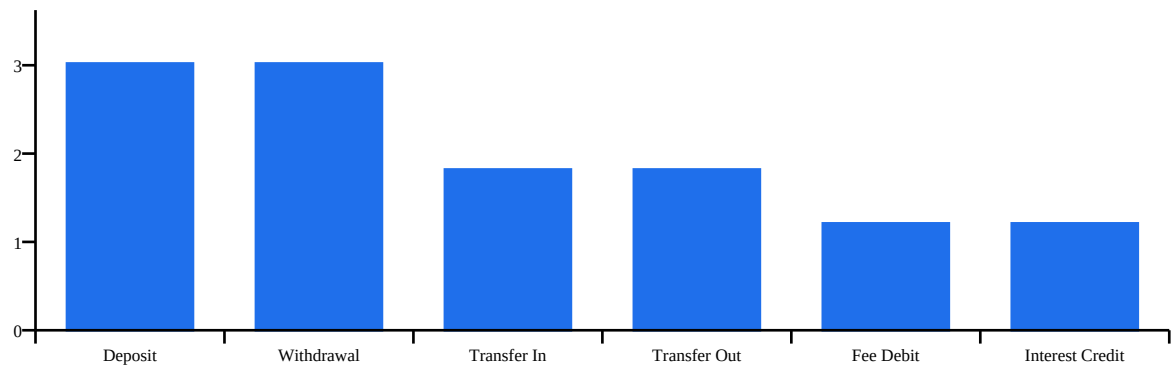
Business implication

Use state, branch, account type and activity status for customer engagement, service-capacity and dormant-account monitoring.

05 — Transaction Analytics

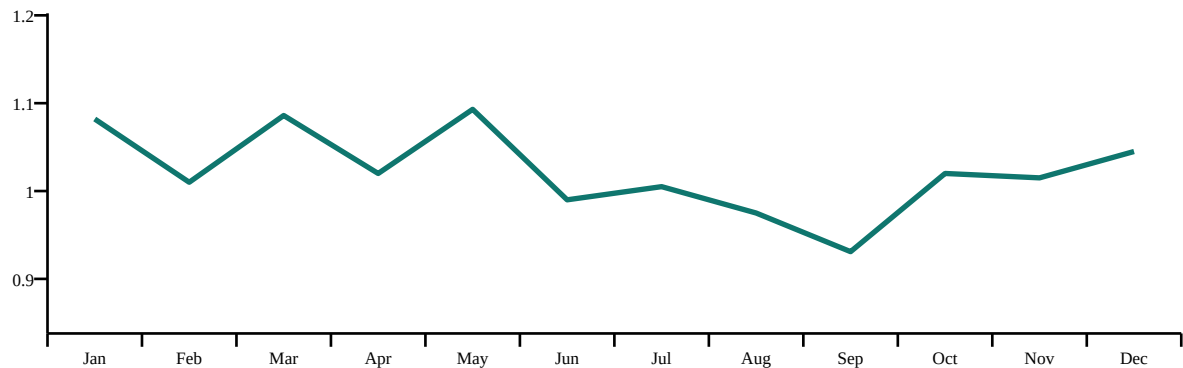
2.0M	■12.10B	■6,048.38	333,776
Core transactions	Core value	Average transaction	Mobile App

Transaction value by type — ■B



Deposit and withdrawal transactions each contribute approximately ■3.02B. Transfer In/Out contribute roughly ■1.82B each; Fee Debit and Interest Credit are around ■1.21B each.

Monthly transaction value — ■B

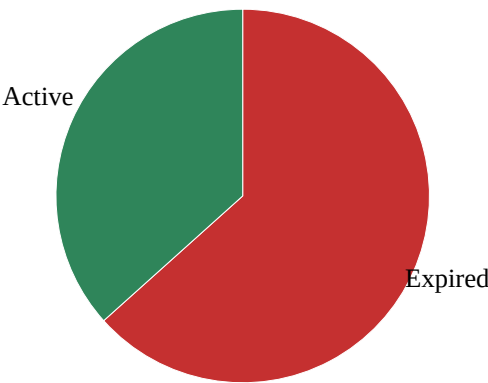


May is highest at approximately ■1.093B; March ■1.086B; January ■1.082B. September is lowest at approximately ■0.931B. The source notes that 2026 is partial.

06 — Cards, Loans & Credit Risk

65K	63.36%	22K	12.0063%
Cards	Classified expired	Loans	Late-payment flags

Card lifecycle classification from notebook expiry logic



41,185 cards (63.36%) are classified as expired and 23,815 (36.64%) as active. Debit cards are the largest card type with 35,700 cards.

Loan repayment metrics

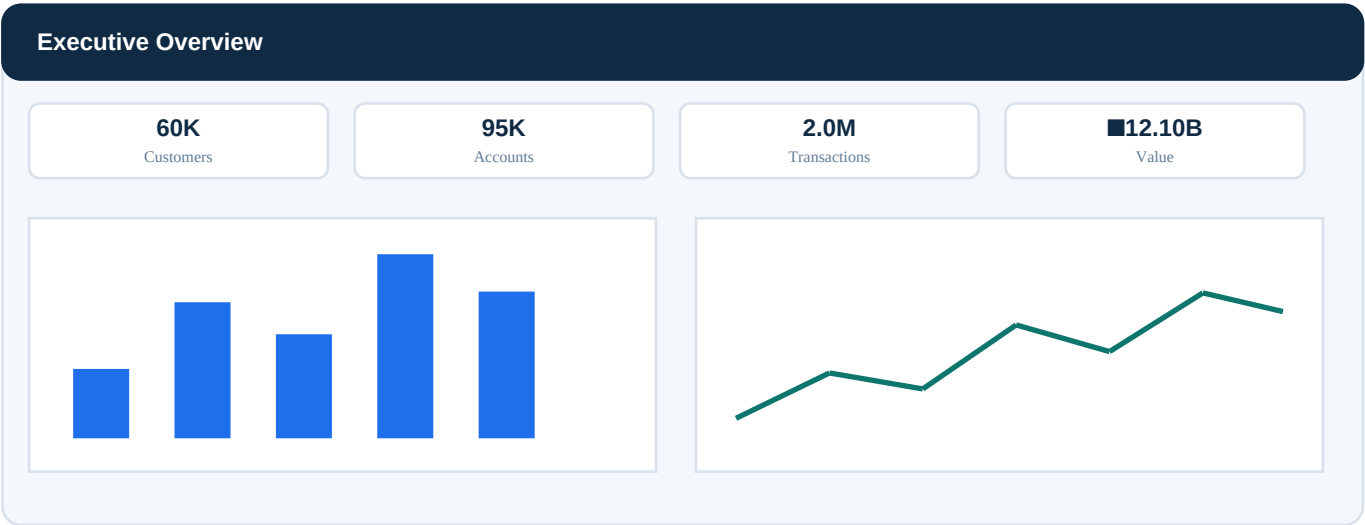
Metric	Value
Loans	22,000
Loan payments	600,000
Late-payment flag rate	12.0063%
Average payment	■10,397.46
Average principal	■8,487.30
Average interest	■1,910.16

Risk-analysis design

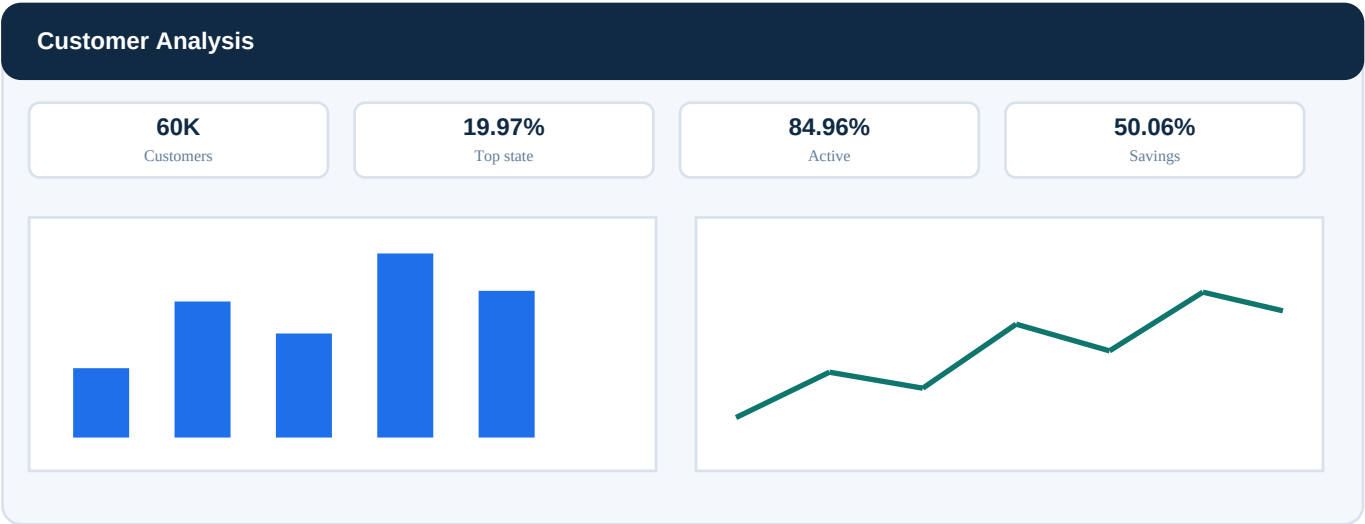
The SQL layer estimates outstanding principal as original loan amount less cumulative principal paid and separately identifies late/overdue patterns. These are descriptive indicators, not causal credit-risk conclusions.

07 — Power BI Dashboard Design

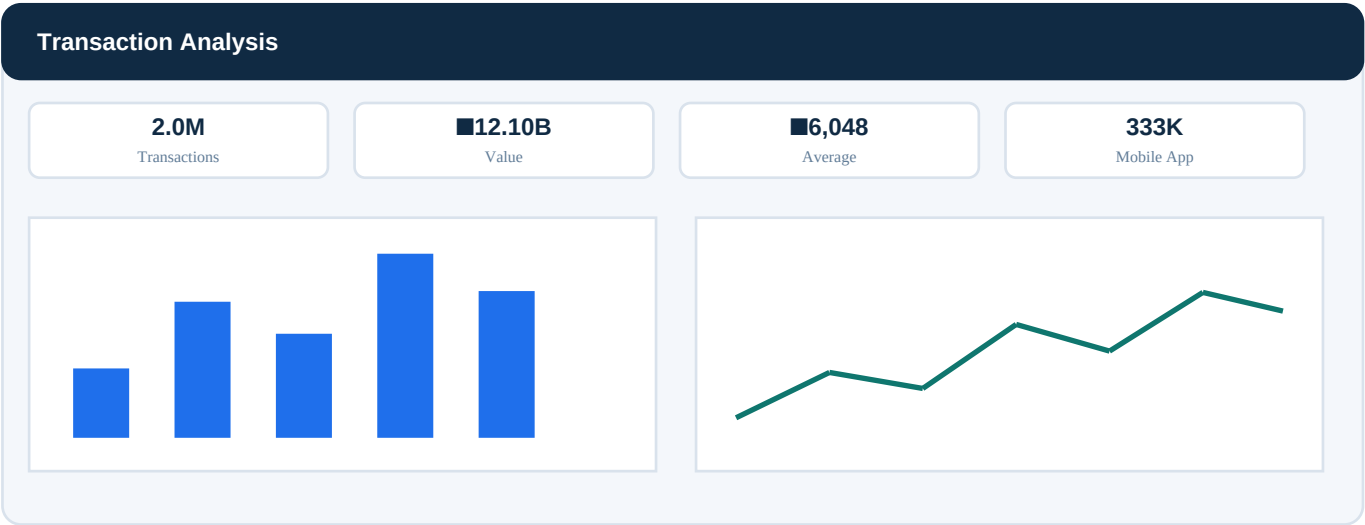
The PBIX is organized into five analytical pages plus a hidden tooltip page. The visual structure is designed around business questions.



01 Executive Overview: bank-wide KPI snapshot and management focus areas.

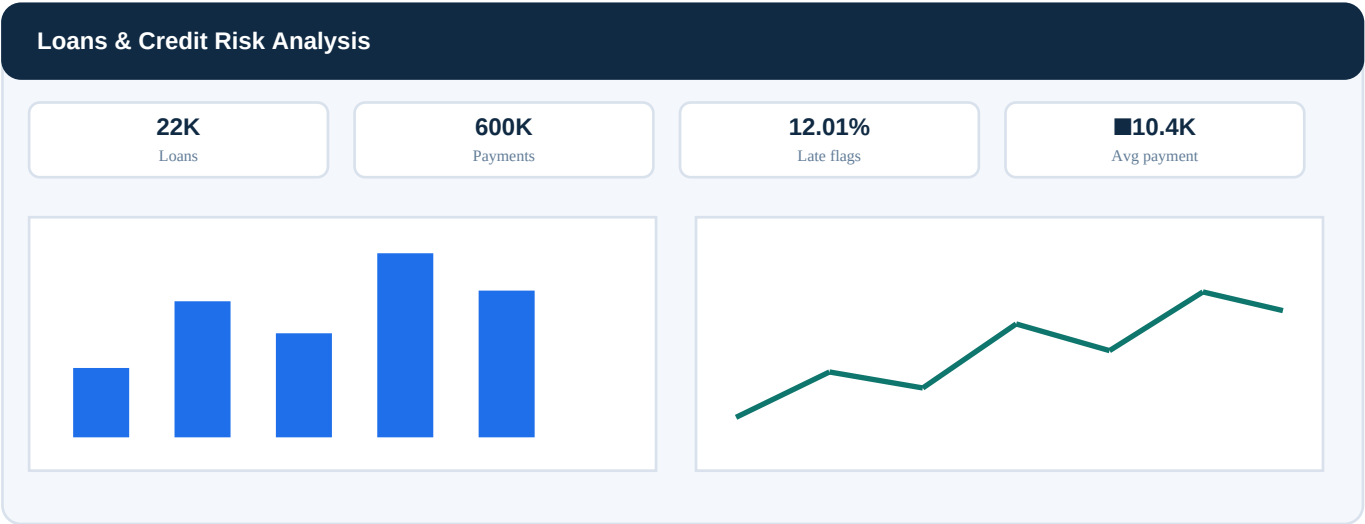


02 Customer Analysis: customer demographics, geography and account relationships.

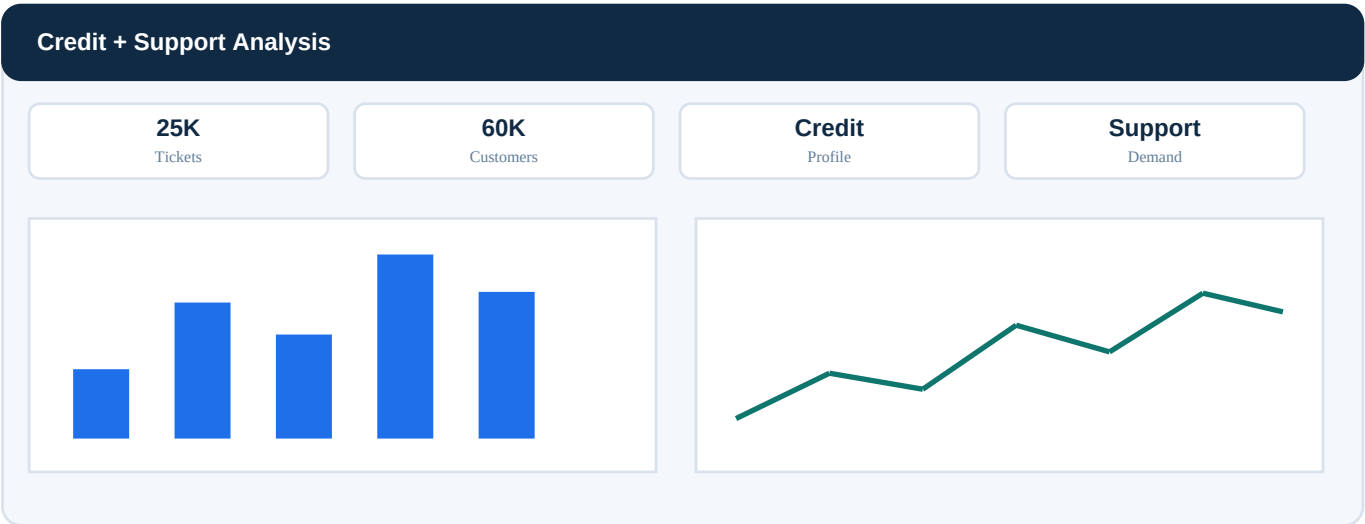


03 Transaction Analysis: volume, value, channels, types and trends.

08 — Dashboard Pages & Recommendations



04 Loans & Credit Risk: lending, repayment and credit-related indicators.



05 Credit + Support: credit characteristics and service operations.

Recommendations

1. Monitor dormant accounts by branch, account type and customer segment.
2. Compare digital channels using both transaction volume and transaction value.
3. Create a card-expiry pipeline by card type and branch.
4. Monitor loan repayment and late-payment patterns by loan type and customer.
5. Use geographic segmentation for customer acquisition and service planning.
6. Combine credit and support indicators for deeper customer-service investigation.

09 — GitHub Packaging & Conclusion

The case study is designed to sit alongside the Power BI file, SQL script and Python notebook in a GitHub repository.

Recommended repository structure

```
banking-transactions-powerbi/  
■■■ README.md  
■■■ CASE_STUDY.md  
■■■ data/README.md  
■■■ python/Banking_Transactions.ipynb  
■■■ sql/banking_transactionsSQL.sql  
■■■ powerbi/banking_transactions.pbix  
■■■ screenshots/  
■■■ documentation/Banking_Transactions_Professional_Case_Study.pdf
```

Skills demonstrated

Data ingestion and profiling • data-quality validation • Python/Pandas EDA • SQL and relational modeling • joins and aggregations • Power BI semantic modeling • DAX and KPI development • dashboard storytelling • business insight generation

Conclusion

The project demonstrates a complete Data Analyst workflow from raw banking data to decision-oriented reporting. Its portfolio value comes from combining technical execution with business interpretation: measurable patterns are connected to practical questions and next-step actions.

Portfolio tagline: Turning multi-domain banking data into actionable business insights with Python, MySQL and Power BI.

Source note: quantified figures are based on the supplied project report and SQL source. 2026 transaction data is partial, and exploratory SQL should be validated before production use.